

GOVIND MITTAL

Senior Data Scientist · AI / LLM / Agentic Systems

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SUMMARY

Senior Data Scientist / AI Engineer with 4+ years designing and shipping **production agentic-AI systems** at enterprise scale (Gartner, Eli Lilly, Citigroup). Specialize in multi-agent orchestration, RAG / knowledge-graph reasoning, LLM fine-tuning, and evaluation. Repeated track record of turning novel agent architectures into measurable, in-production impact in high-stakes finance and operations.

EXPERIENCE

Gartner — Senior Scientist, AI · Feb 2026 – Present · Bengaluru, India

- **GraphGOAT**: re-architected Gartner's agentic tech (GOAT) around a **single knowledge graph** — unifying the data behind post-call follow-ups, quarterly business reviews, and asset-intelligence agents so the agent only writes a graph query and summarizes, instead of doing multi-step reasoning itself. Cut context ~**200–250K** → **<15K tokens** per query, per-query reasoning **10–15 steps** → **2–3**, and agentic errors ~**20%**, while opening far wider data access — across **30–50K calls/month**.
- **Local LLM migration**: fine-tuned and deployed a **Gemma-4 12B** model on AWS SageMaker (L4 GPU), replacing recurring OpenAI API cost with local inference. With **multi-token prediction** and **TurboQuant** near-lossless 4-bit KV-cache quantization, cut decode latency ~**50%** and lifted throughput ~**50%** — matching cloud-API speed at no per-call cost.
- **Agent performance & evaluation engineering**: TOON+Markdown I/O (**–22% latency, –35% tokens**), prompt caching, model routing, DSPy for model-invariant prompts, and RL-based feedback with LLM eval suites.

Eli Lilly — Senior Scientist, AI (Tech@Lilly) · Apr 2025 – Nov 2025 · Bengaluru, India

- **Nexify**: built an **ambient multi-agent system** automating the full data-science lifecycle — analysis, modeling, forecasting, validation, review, research, and writing — compressing **question** → **insight from weeks to minutes**. **Fully retrained** a Qwen-2.5 Coder model for the pipeline (re-training all attention and MLP layers from the pretrained checkpoint) via a "workflow-as-code" approach.
- **ARTIE (Artificial Teams for Rapid & Enhanced Insights)**: a **JSON-schema-driven agent builder** — the orchestrating agent emits only a JSON schema and the system self-constructs and executes the LangGraph workflow on the fly. Scaled **300** → **900+ users**.
- **CFO Notes & VizSnap**: architected CFO Notes, a multi-agent RAG platform (LangGraph + FastAPI) for financial variance reporting across **20 products / 18 markets (1 day → 15 min)**; flagship tool for Lilly USA Finance); and VizSnap, an MCP-based agentic visualization system (**900+ users, 10K+ queries/mo**; fine-tuned Qwen2.5-VL via RL), evaluated with LangSmith.

Citigroup — Senior Data Scientist (Business Analyst → Data Scientist → Sr. DS) · Jul 2022 – Mar 2025 · Bengaluru, India

- **Project Luna (Graph-RAG IVR containment)**: a knowledge graph of call reasons and resolutions with an agent that infers the customer's *underlying* need (e.g., "can't transact on my card" → resolve via phone-banking) to avoid unnecessary human transfers. Deployed to production; ~**\$2.4M/month** saved in the Citibank North America pilot.
- **Call-volume forecasting**: hierarchical time-series with N-Beats + Transformer ensembles for capacity planning; accuracy **81%** → **91.8%**.
- **Deposit-rate prediction**: Random-Forest + Transformer ensemble for interest-rate optimization across **2,500+ institutional clients**; prediction error down to ~**11 bps**.
- **Fraud & compliance**: a first-party fraud anomaly-detection system saving ~**\$6.6M/year**; and a multi-agent compliance-QC framework (analyst time ~**52%**).

SKILLS

Agentic AI & LLMs: LangGraph, LangChain, MCP, LangSmith, DSPy, RAG / Graph-RAG, multi-agent orchestration, knowledge graphs, fine-tuning (LoRA/QLoRA, full retraining, Unsloth), RL (Agent Lightning), prompt/context engineering, evals & observability, safety/guardrails.

ML & Quantitative: time-series forecasting (N-Beats, hierarchical), anomaly detection, Transformer ensembles, reinforcement learning, econometrics, Bayesian methods, causal inference, hypothesis testing, regression & factor modeling, PyTorch, TensorFlow, explainable AI (SHAP/LIME).

Knowledge Graphs & Retrieval: Neo4j, AWS Neptune, Pinecone, Weaviate, Chroma, FAISS, Elasticsearch, semantic search, entity linking.

Engineering & Cloud: Python, SQL, R, FastAPI, Docker, Kubernetes, CI/CD, MLOps, AWS (SageMaker), GCP, Azure, Snowflake, PostgreSQL, PySpark.

SELECTED PROJECTS

- **StartBrief** — client-call intelligence platform (built for Gartner): extracts reliable, structured insight from large volumes of client calls using a LangGraph + RAG pipeline.
- **AgentFlow** — an "agent factory": hand it a task and it assembles the specialized sub-agents (and the code to run them) needed to execute it, wiring them into a working LangGraph flow.
- **AI Economist** — RL-based policy-optimization agent extending Salesforce Research's AI Economist to search for optimal tax / economic policy in simulated economies.
- **AI Shayar** — an Indic-language poetry LLM: fine-tuned Gemma-2 (QLoRA via Unsloth) on Ghalib's Urdu/Hindi verse to generate original shayari.
- **Mechanistic Interpretability** — reproduced Anthropic's attribution-graph / circuit-tracing approach for visualizing transformer neuron pathways and interpretability.

EDUCATION

MSc Economics — Gokhale Institute of Politics & Economics, Pune · 2020 – 2022

Batch Prize — highest scores in Econometrics, Business Analytics & Linear Economics.

BSc Economics & Statistics — NMIMS University, Mumbai · 2015 – 2018

Best Paper Presentation & Budding Economist awards.

CERTIFICATIONS

LangChain Academy — Deep Agents, Ambient Agents, Introduction to LangGraph, Agent Observability & Evaluations · Hugging Face — AI Agents · Kaggle — Intro to Deep Learning.